Neutral-current four-bottom production

Unpolarized FCC-ue near 1 TeV | Three completed MG5_aMC@NLO runs | INTERNAL VALIDATION

Beam energies: E(μ+) = 2000 GeV; E(e−) = 120 GeV; √s = 979.796 GeV. MadGraph5_aMC@NLO 3.6.4, default restricted SM, tree level; 30,000 generated events for each channel.

Cross sections from the archived run reports

Process	Before b-cuts (fb)	After b-cuts (fb)	Approx. MC ± (fb)	Acceptance
NC-HH · 4 diagrams	0.00643724	0.00638240	0.00000990	99.1482%
NC-bbH · 86 diagrams	0.02040127	0.01292500	0.00005939	63.3539%
Complete NC-4b · 1,902 diagrams	0.52349770	0.00771287	0.00036434	1.4733%

“Before b-cuts” is not fully inclusive: all three calculations already impose pT(μ+ out), pT(e− out) > 1 GeV as a forward-scattering regulator.

Common fiducial definition and branching fractions

Four final-state bottom partons in the laboratory frame; |η(b)| < 5; two disjoint opposite-charge b-bbar pairs each with invariant mass in [110, 130] GeV. There is no minimum b-parton transverse-momentum cut, jet clustering, showering, or detector response.

Model-card BR(H → b-bbar) = 0.8446400227 (not 0.582). The NC-HH rate includes BR^2, the NC-bbH rate includes BR, and the complete NC-4b rate contains no extra branching factor. The separate channels overlap and must not be added to the complete result.

Priority issue: full NC-4b rate requires validation

Full NC-4b fiducial: 0.00771287 fb; separate on-shell NC-bbH fiducial: 0.01292500 fb. The full result is only 59.7% of the bbH result. The full sample has 442 accepted events / 30,000 (1.4733%); the quoted approximate MC error is 4.72% relative. This discrepancy is not explained by these files and must not be attributed to interference without a dedicated check.

All three numerical integrations and subsequent analyses completed. Status remains ESTIMATE_NOT_CONVERGENCE_VALIDATED; none is yet a precision prediction.

Audit findings and follow-up checks

What the archived evidence verifies, and what it cannot yet establish

Verified from the supplied ZIP

1. The archive contains all three individual run reports, process cards, identical SM parameter cards, and completed generator/analysis logs. All 45 entries listed in the package SHA-256 manifest match their hashes.

2. The complete NC-4b channel uses mu+ e− → mu+ e− b bbar b bbar with QCD ≤ 2 and QED ≤ 6; generator output reports 1,902 tree diagrams. The associated-Higgs channel has 86 diagrams, and the HH channel has four.

3. All 196 normalized run-card entries have the same values across the three runs; beam energies, polarization, generator cuts, fixed reference scale, and selection metadata match.

4. Reported fiducial rates reproduce $\sigma(\mathrm{MG5}) \times 1000 \times \mathrm{BR}^n \times$ acceptance, with $n = 2, 1, 0$ for HH, bbH, full-4b. The archived approximate MC errors are arithmetically consistent.

Interpretation and limitations

The bbH production sample can contain contributions with a second resonant Higgs decaying to the prompt b-bbar pair, so it is not guaranteed to be an exclusive single-Higgs contribution. All amplitudes within the complete four-bottom process are coherently combined; interference and resonance treatment complicate comparisons to factorized on-shell samples.

The quoted uncertainties combine the generator integration and event-selection uncertainties in quadrature, neglecting covariance and all theoretical/systematic effects. The source ZIP deliberately excludes LHE event files; consequently, independent event-by-event validation cannot be completed using the archive alone.

The original configuration.json has a stale template field “PREPARED_ONLY...” even though the separate manifest.json and generator reports unequivocally document completed runs. The diagnostic package leaves original archived files unchanged.

Recommended scientific validation

1. Replay the complete-4b LHE with an independent invariant-mass/pairing implementation; its fiducial count should reproduce 442. Inspect the single- and double-Higgs peaks around 125 GeV.

2. Inspect bbH prompt b-bbar invariant mass to measure the size of a second resonant Higgs contribution, and avoid labelling the bbH sample as an exclusive non-HH background.

3. Repeat the full 4b fiducial prediction with an independent integration seed and resonance-aware phase-space strategy or suitable fiducial generation; compare both results and convergence. Use CERN batch resources for intensive runs.

4. Check outgoing-lepton regulator sensitivity and eventual experimental acceptance before combining charged- and neutral-current samples in an observable selection.

Sharing recommendation

The NC-HH and NC-bbH on-shell estimates can be shared with Krzysztof as preliminary, with their definitions and caveats. The complete NC-4b number should be labelled under validation until its resonance sampling and comparison to the on-shell rates are checked. The earlier CC diagram request remains separate and is not covered by this ZIP.